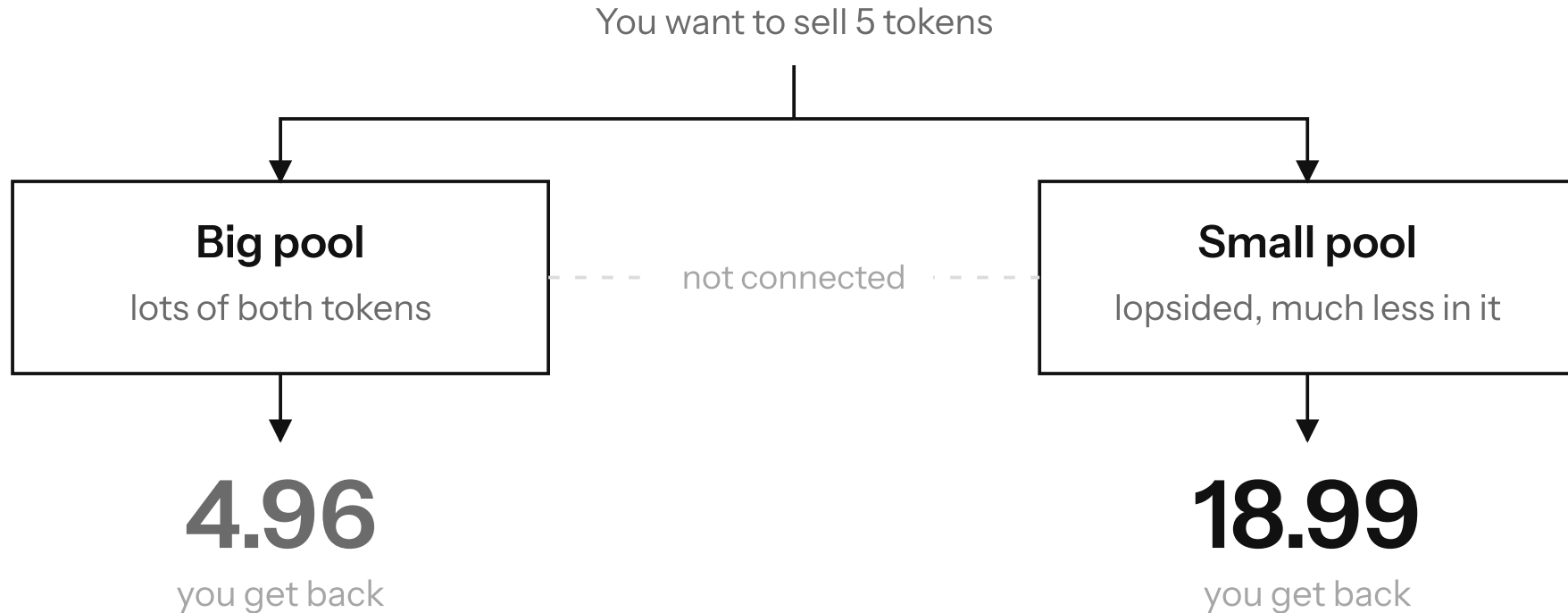


Knot

Two pools holding the same tokens
should not quote two different prices.

A Uniswap v4 hook. Live on Unichain Sepolia.

The same trade. Two prices.



The small pool pays four times more. That comes out of its own investors.

Our fix: ask for the price twice.

Price one

What this pool alone would pay you

Price two

What all the pools together would pay you

You get

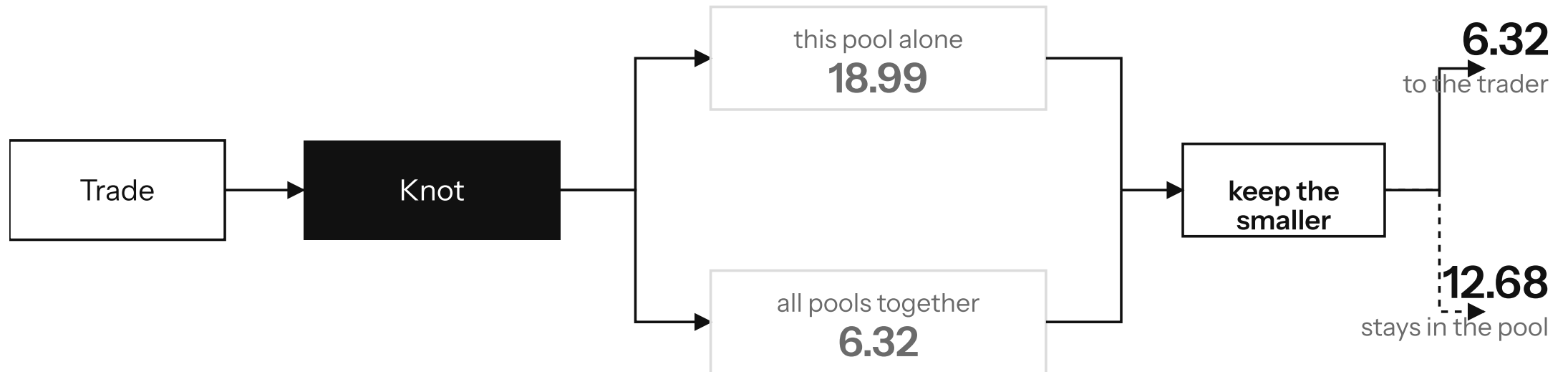
Whichever of the two is worse for you

The rest

Stays in the pool, for the people who funded it

No outside price feed. No auction. No guessing who is a good or bad trader.

What happens during one trade.

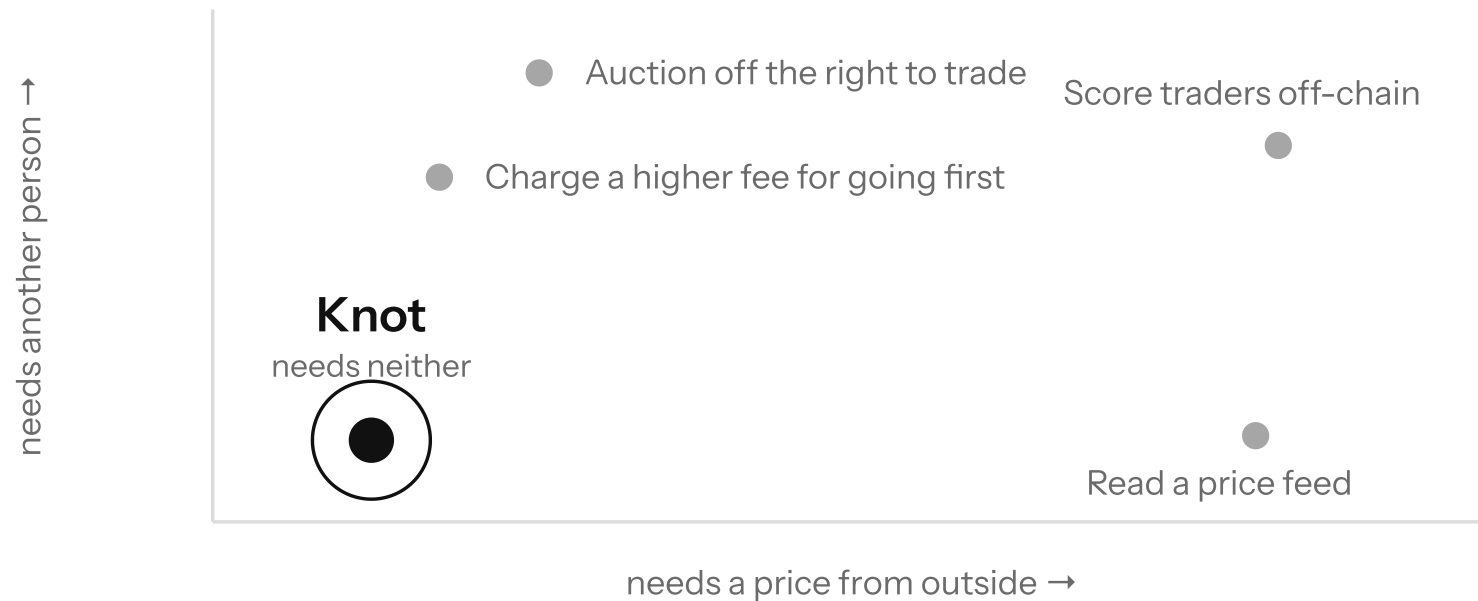


The whole rule, in real code.

```
if (exactInput) {  
    localQuote      = KnotMath.amountOut(amount, localIn, localOut, feeNum, feeDen);  
    aggregateQuote = KnotMath.amountOut(amount, aggIn,  aggOut,  feeNum, feeDen);  
    knotQuote = localQuote < aggregateQuote ? localQuote : aggregateQuote;  
} else {  
    localQuote      = KnotMath.amountIn(amount, localIn, localOut, feeNum, feeDen);  
    aggregateQuote = KnotMath.amountIn(amount, aggIn,  aggOut,  feeNum, feeDen);  
    knotQuote = localQuote > aggregateQuote ? localQuote : aggregateQuote;  
}
```

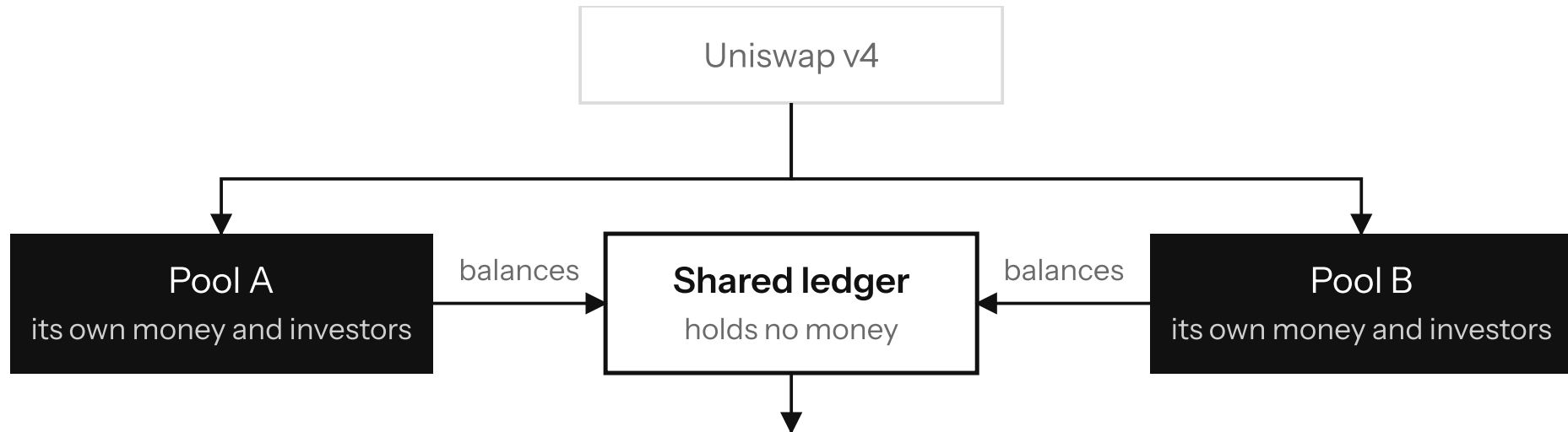
Two sums, then pick one. Nothing here looks at who is trading.

We check the pool, not the trader.



Nobody has built this before. We only use numbers the pools already hold.

Pools share numbers, not money.



the two prices, worked out in the same instant

The check sits inside the pool, so no app can route around it.

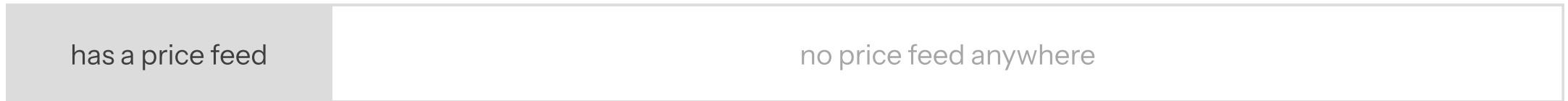
New money cannot move the price.



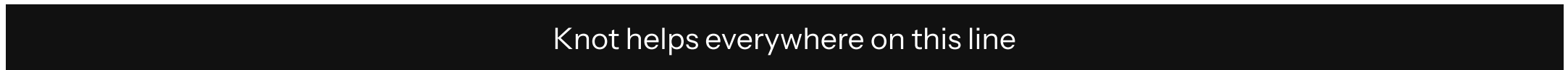
Otherwise someone could add money, bend the price, take it, and walk away.

Works on any pair, even a new one.

Every pair of tokens on Uniswap



Anything that reads a price feed only helps here ↑



drawn to make the point, not to scale

Small and new pairs are where people lose the most money.

Finished, and tested hard.

182

tests, all passing

99%

of our code covered by
them

163,840

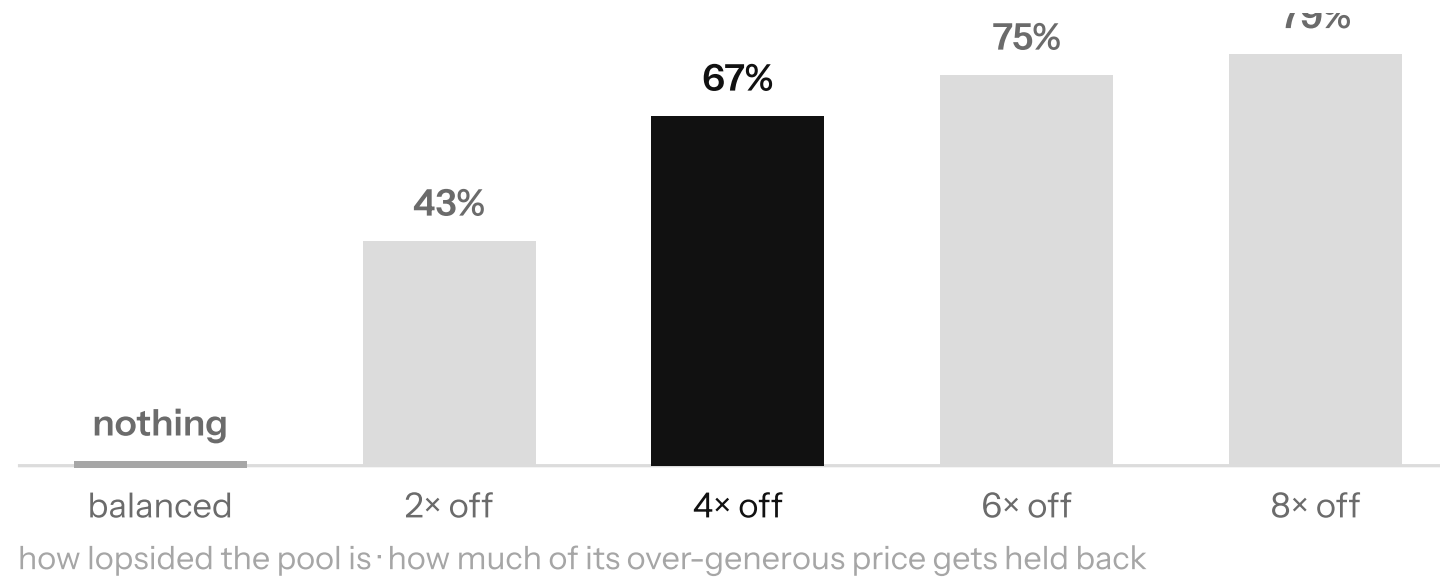
random attack sequences
run

0

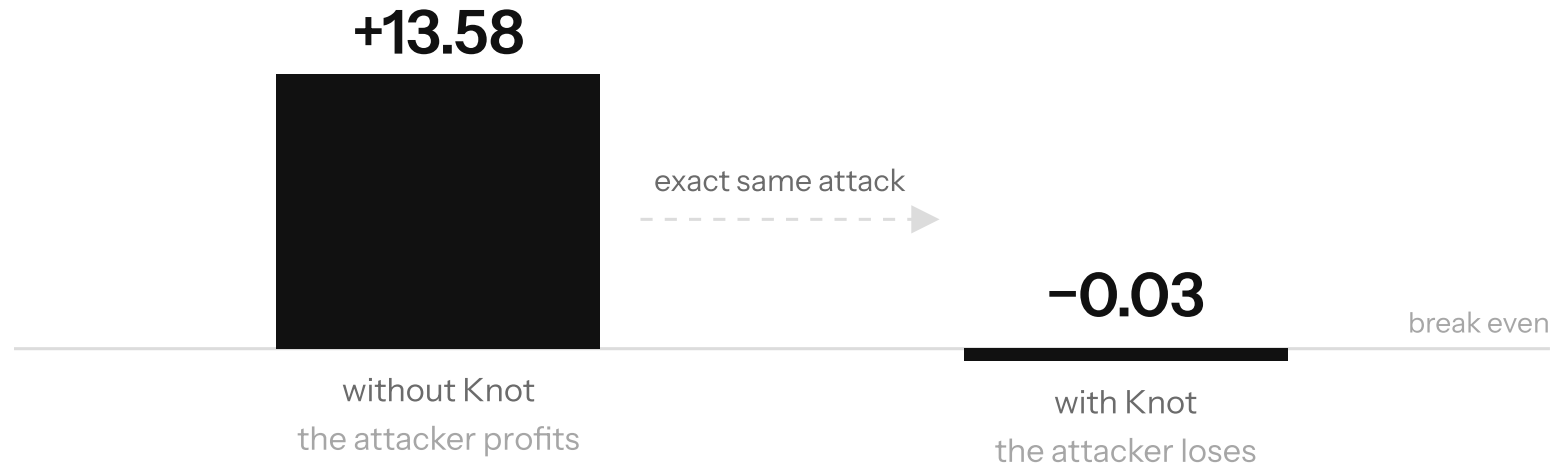
failures in any of them

Running on a public test network. Anyone can verify the code we published.

A fair pool gives up nothing.



An attack that paid now loses.



We tried every size, from tiny to large. It never pays.

Checked against 67,743 real trades.

We asked

Which trades actually cost pool investors money?

We found

The ones getting a better price here than at a sibling pool

Which means

That is exactly the kind of trade our rule holds back

Real trades from Base, not a simulation. We measured it, we did not guess.

Nobody has connected pools this way before.

It needs no price feed, so it works on any pair.

It is finished, tested and running on a live network.

It cannot be dodged, because it lives inside the pool.

And it is simple enough to explain in one sentence.

**Two pools. Same
tokens. One fair price.**

knot-hook-web.vercel.app · github.com/Hijanhv/knot-hook